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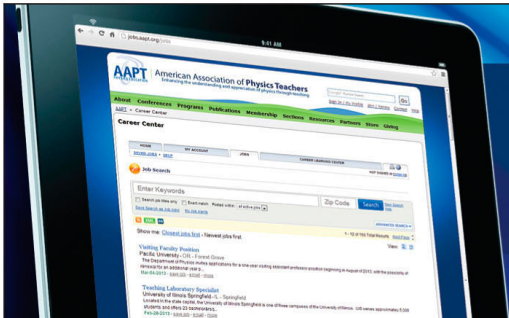
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A brief primer on tamped fission-bomb cores

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A previously published discussion of the physics of fission-bomb cores is extended to incorporate the effects of a surrounding tamper/reflector. I examine the effect of a tamper of finite thickness on the calculation of critical mass in contrast to the infinite-thickness approximation usually invoked. An infinite-thickness tamper reduces the critical mass by a factor of eight; a more realistic estimate for a tamper of nonzero mass is a factor of about two. An approximate model of the Hiroshima *Little Boy* bomb results in a predicted explosive yield of about 13 kilotons, in good agreement with the actual value of approximately 12 kilotons, although this agreement might be fortuitous. A spreadsheet for estimating the critical masses of tamped and untamped cores and expected bomb yields and efficiencies is available. © 2009 American Association of Physics Teachers.
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I. INTRODUCTION

In a previous paper I examined the conditions necessary for a fission weapon and its expected efficiency for an untamped spherical core of fissile material.¹ The present paper extends the considerations to include tamped cores. My goal is to offer a more complete treatment of fission-bomb physics and to clarify the extent to which a tamper acts to reduce the critical mass. References 2 and 3 consider the case of an infinitely thick tamper, which reduces the critical radius by a factor of two in comparison to the case of no tamper. I allow the tamper to have any thickness by applying the relevant boundary conditions.^{2,3}

A tamper increases the efficiency (and hence the yield) of a weapon due to two effects. By briefly retarding the expansion of the core, the tamper causes the core to remain at a higher density for longer than it would have otherwise. This delay is beneficial because the efficiency is proportional to the average density of the core. The tamper also serves to reflect neutrons back into the core, effectively decreasing the loss of fission-causing neutrons from within the core to the surroundings. In modern weapons engineering parlance, a tamper is known as a reflector. For consistency with Refs. 2 and 3, I use the historical terminology. The retardation effect is difficult to model easily and is not treated here, but the reflection effect can be modeled by straightforward modifications of the boundary conditions invoked in Ref. 1 to derive the condition for untamped criticality.

Before proceeding, an important limitation of the model discussed in Ref. 1 needs to be clarified. As in Ref. 2, I model the motion of neutrons in the bomb core by a diffusion equation. A diffusion approach is appropriate if neutron scattering is isotropic. Even if the scattering is anisotropic, a diffusion approach will be reasonable if the neutrons suffer a large enough number of scatterings so as to effectively cancel nonisotropic angular effects. Neither of these conditions is satisfied in a uranium core: fast neutrons elastically scattering against uranium show a strong forward-peaked effect, and the mean free path of a fast neutron in a U-235 core is only about half of the 8.3 cm untamped critical radius (see Table I in Ref. 1). I adopt a diffusion approach for several reasons. Because much of the physics in this area remains classified or is not easily accessible, we are forced to settle for an approximate model. Diffusion theory has the advantage of being analytically tractable, and, despite its limita-

tions, the critical radii predicted by diffusion theory and by a more exact treatment agree to within about 5% for the range of fissility parameters of interest here.⁴ Some further approximations are discussed in Ref. 5.

II. UNTAMPED CRITICALITY

As described in Ref. 1, the diffusion equation for the space- and time-dependence of the number of neutrons within a bomb core is given by

$$\frac{\partial N}{\partial t} = \frac{v_n}{\lambda_f^{core}}(\nu - 1)N + \frac{\lambda_t^{core}v_n}{3}(\nabla^2 N), \quad (1)$$

where N is the number density of neutrons, v_n is the average neutron velocity, ν is the number of secondary neutrons produced per fission, and λ_f^{core} and λ_t^{core} are the fission and transport mean free paths for neutrons in the core, respectively. These mean free paths are given by

$$\lambda_f^{core} = \frac{1}{\sigma_f n} \quad (2)$$

and

$$\lambda_t^{core} = \frac{1}{\sigma_t n}, \quad (3)$$

where n is the nuclear number density and σ_f and σ_t are the fission and transport cross-sections. If the neutron scattering is isotropic as is assumed here, the transport cross-section is given by the sum of the fission and elastic-scattering cross-sections:⁵

$$\sigma_t = \sigma_f + \sigma_{el}. \quad (4)$$

An important quantity is the average time τ that a neutron travels before causing a fission:

$$\tau = \lambda_f^{core} / v_n. \quad (5)$$

Equation (1) can be solved by separation of variables. If the separation constant is written as α/τ , the solution is

$$N_{core}(r, t) = N_0 e^{(\alpha/\tau)t} \left[\frac{\sin(r/d_{core})}{r} \right], \quad (6)$$

where

$$d_{core} = \sqrt{\frac{\lambda_f^{core} \lambda_t^{core}}{3(-\alpha + \nu - 1)}}. \quad (7)$$

N_o is the initial neutron number density, which would be set by the initiating device that starts the chain reaction. The quantity d_{core} has units of length and sets the size of the critical radius [see Eq. (9)].

To determine the critical radius R_C , we need a boundary condition to apply to Eq. (6). As explained in Ref. 1, this condition takes the form

$$N(R_C) = -\frac{2\lambda_t^{core}}{3} \left(\frac{\partial N}{\partial r} \right)_{R_C}. \quad (8)$$

This condition arises in the derivation of the diffusion equation by considering the net rate of diffusion of neutrons through a spherical surface. If the surface is the outer edge of the core, we require that the neutrons that have passed through the surface cannot be reflected back from the outside. The application of this requirement to Eq. (6) gives the condition for untamped criticality as

$$(R_C/d_{core}) \cot(R_C/d_{core}) + (3d_{core}/2\lambda_t^{core})(R_C/d_{core}) - 1 = 0. \quad (9)$$

If $\alpha=0$, then the neutron number density neither increases nor decreases in time. The time when this state first occurs during assembly of subcritical core components into a supercritical mass is known as *first criticality*. I also refer to $\alpha=0$ as threshold criticality. In this case, d_{core} is completely determined by the fissility parameters, and Eq. (9) can be solved numerically for the bare critical radius, $R_C = R_{bare}^{thresh}$. (An untamped core is known as a bare core.) As explained in Ref. 1, if we start with a core of radius $R > R_{bare}^{thresh}$ and use R in place of R_C in Eq. (9), then Eq. (9) can be solved for α (through d_{core}), and we find that $\alpha > 0$, which means that the neutron density grows exponentially in time, a condition known as “supercriticality.”

If we begin with a core with $R > R_{bare}^{thresh}$, the expansion and consequent decreasing density of the core due to gas and/or radiation pressure will lead, after a short time, to the loss of criticality; that is, α will decrease from its initial value to zero at a time known as *second criticality*. Hence, α in Eqs. (6) and (7) is a function of time. To avoid having to treat this complication, I will take α to be an effective α given by the average of these two values, $\alpha_{initial}/2$. This approximation is not strictly valid because the core expands exponentially rather than linearly in time, but since we are using an approximate diffusion model and will need to introduce some further approximations later, it is not worthwhile to attempt a more rigorous analysis.

Imagine a core of mass m , and let $C = m/m_C$, where m_C is the critical mass corresponding to R_{bare}^{thresh} . If the efficiency is defined as the ratio of the total energy released up to the time of second criticality to the total which would be released if every nucleus in the core were to fission, we find

$$\text{Efficiency} = \frac{(\Delta r) \alpha^2 \langle \rho r \rangle}{3 \gamma m \tau^2 E_f}. \quad (10)$$

The terms Δr and $\langle \rho r \rangle$ in Eq. (10) relate to the fact that as the core expands, its radius r increases while its density ρ decreases. In Ref. 1, it is explained how the condition of criticality is dictated by the product ρr , such that as long as

$\rho r > \rho_o R_{bare}^{thresh}$, where ρ_o is the initial density of the material, criticality will hold. (If it is desired to model an implosion weapon, replace ρ_o by the density of the core when fissions are initiated.) If the core material starts at a normal density, its initial radius will be $r_i = C^{1/3} R_{bare}^{thresh}$. As it expands, its density at radius r is $\rho = C \rho_o (R_{bare}^{thresh}/r)^3$. Second criticality occurs when $\rho r_{crit} = \rho_o R_{bare}^{thresh}$ or after eliminating ρ , $r_{crit} = C^{1/2} R_{bare}^{thresh}$. The range Δr of the radius over which criticality holds becomes

$$\Delta r = (r_{crit} - r_i) = (C^{1/2} - C^{1/3}) R_{bare}^{thresh}. \quad (11)$$

The quantity $\langle \rho r \rangle$ in Eq. (10) is the average of the product of the core's radius and density. This product is a function of time but can be treated by a convenient approximation for any reasonable value of C . From the discussion in the previous paragraph, $(\rho r)_{initial} = C^{1/3} (\rho_o R_{bare}^{thresh})$. Because at second criticality $(\rho r)_{crit} = (\rho_o R_{bare}^{thresh})$, we have $(\rho r)_{crit}/(\rho r)_{init} = C^{-1/3}$. For $C=2$, for example, we obtain $(\rho r)_{crit}/(\rho r)_{init} \approx 0.79$; that is, $(\rho r)_{crit}$ and $(\rho r)_{initial}$ do not differ greatly. Hence, I assume that the product $\langle \rho r \rangle$ in Eq. (10) can be approximated by the average of the values of ρr at the initial and second criticality conditions:

$$\langle \rho r \rangle = \frac{1}{2} (1 + C^{1/3}) \rho_o R_{bare}^{thresh}. \quad (12)$$

The factor E_f in Eq. (10) is the average energy released per fission (typically ~ 180 MeV), and γ depends on whether gas pressure ($\gamma=2/3$) or radiation pressure ($\gamma=1/3$) is dominant. The choice of an appropriate value for γ is not straightforward. One approach is to consider the average energy released by fissions per nucleus of core material to the time of second criticality, which can be shown to be equal to the efficiency times E_f . If the efficiency is very low, say 0.1% for the sake of argument, this average energy would be 180 keV for $E_f=180$ MeV. The radiation and classical particle gas pressures are given by $P_{rad} = (8\pi^5 k^4 / 45 c^3 h^3) T^4$ and $P_{gas} = (10^6 \rho N_A k / A) T$, where ρ and A are the mass density and atomic weight of the gaseous material in g/cm³ and g/mol, respectively. With $E = 3kT/2$, these expressions show that the radiation pressure will dominate over the gas pressure for $E > 4145(\rho/A)^{1/3}$ eV. For U-235 nuclei with $(\rho, A) = (18.71, 235)$, this condition corresponds to 1.8 keV. This figure for the density of U-235 arises from taking $(235/238)$ of the density of natural uranium, 18.95 g/cm³.⁶ On this basis, it would seem reasonable to adopt $\gamma=1/3$, even if the bomb efficiency is extremely low. However, we expect that a significant fraction of the available energy will likely be carried away by the many free electrons resulting from ionization. Results given in Sec. IV show that $\gamma=2/3$ is probably the more realistic choice.

III. EFFECTS OF A TAMPER

Imagine that the fissile core is surrounded by a snugly-fitting spherical tamper of outer radius R_{tamp} . The mass of such a core is known as a “reflected mass” or “tamped mass.” Neutrons that diffuse through the surface of the core will enter the tamper. To describe the behavior of neutrons in the tamper we can use Eq. (1) without the term corresponding to the production of neutrons:

$$\frac{\partial N_{tamp}}{\partial t} = \frac{\lambda_t^{tamp} v_n}{3} (\nabla^2 N_{tamp}), \quad (13)$$

where λ_t^{tamp} is the mean free path for neutrons in the tamper. For simplicity, I will assume that neutrons have the same speed within the core and tamper materials. I also assume that the tamper does not absorb neutrons.

Equation (13) can also be solved by separation of variables. We write the separation constant as δ/τ , where τ is the mean travel time for neutrons between fissions in the core, Eq. (5). This choice is advantageous in that the neutron velocity v_n cancels. The physical interpretation of δ is analogous to that of α in Eq. (6), and it dictates whether the neutron density in the tamper is exponentially growing ($\delta > 0$), decaying ($\delta < 0$), or is constant in time ($\delta = 0$), corresponding to first or second criticality in analogy to $\alpha = 0$. The solution of Eq. (13) depends on the sign of δ ; the situations of practical interest are those with $\delta \geq 0$, in which case

$$N_{tamp} = e^{(\delta/\tau)t} \begin{cases} \left[\frac{A}{r} + B \right] & (\delta = 0) \\ \left[A \frac{e^{r/d_{tamp}}}{r} + B \frac{e^{-r/d_{tamp}}}{r} \right] & (\delta > 0), \end{cases} \quad (14)$$

where A and B are constants of integration, and

$$d_{tamp} = \sqrt{\frac{\lambda_t^{tamp} \lambda_t^{core}}{3\delta}}. \quad (15)$$

The neutron density in the core is described by Eqs. (6) and (7), and the density in the tamper is described by Eqs. (14) and (15). What boundary conditions apply? If no neutrons are created or lost at the core/tamper interface, then the density and flux of neutrons there must be continuous. Also, if there is no backflow of neutrons from the outside into the tamper, we require at the outer edge of the tamper a condition analogous to Eq. (8). If the core has radius R_{core} , these conditions can be written as

$$N_{core}(R_{core}) = N_{tamp}(R_{core}), \quad (16)$$

$$\lambda_t^{core} \left(\frac{\partial N_{core}}{\partial r} \right)_{R_{core}} = \lambda_t^{tamp} \left(\frac{\partial N_{tamp}}{\partial r} \right)_{R_{core}}, \quad (17)$$

and

$$N_{tamp}(R_{tamp}) = -\frac{2}{3} \lambda_t^{tamp} \left(\frac{\partial N_{tamp}}{\partial r} \right)_{R_{tamp}}. \quad (18)$$

Equation (17) expresses the continuity of the neutron flux and incorporates the reflective effect of the tamper through λ_t^{tamp} . If the neutron velocity is different in the two materials, then each side of Eq. (17) must be multiplied by the velocity in the relevant material; see Eq. (2) of Ref. 3.

If we apply Eqs. (16)–(18) to Eqs. (6) and (14), we find that $\alpha = \delta$, and we obtain the criticality expressions:

$$\begin{aligned} & \left[1 + \frac{2R_{tamp}^{thresh} \lambda_t^{tamp}}{3R_{tamp}^2} - \frac{R_{tamp}^{thresh}}{R_{tamp}} \right] \left[\left(\frac{R_{tamp}^{thresh}}{d_{core}} \right) \cot \left(\frac{R_{tamp}^{thresh}}{d_{core}} \right) - 1 \right] \\ & + \frac{\lambda_t^{tamp}}{\lambda_t^{core}} = 0, \quad (\delta = 0) \end{aligned} \quad (19)$$

and

$$\begin{aligned} & e^{2(x_{ct}-x_t)} \left[\frac{x_c \cot x_c - 1 - \lambda(x_{ct}-1)}{R_{tamp} + 2\lambda_t^{tamp}(x_t-1)/3} \right] \\ & = \left[\frac{x_c \cot x_c - 1 + \lambda(x_{ct}+1)}{R_{tamp} - 2\lambda_t^{tamp}(x_t+1)/3} \right] \quad (\delta > 0), \end{aligned} \quad (20)$$

where

$$\left. \begin{aligned} x_{ct} &= R_{core}/d_{tamp} \\ x_c &= R_{core}/d_{core} \\ x_t &= R_{tamp}/d_{tamp} \\ \lambda &= \lambda_t^{tamp}/\lambda_t^{core} \end{aligned} \right\}. \quad (21)$$

Equation (19) corresponds to tampered first (threshold) criticality, where $\alpha = \delta = 0$. Once the values for the d 's and λ 's are given, the only unknown is R_{tamp}^{thresh} . To use Eq. (20), we must first decide on the number C of tampered threshold critical masses of the core and set $R_{core} = C^{1/3} R_{tamp}^{thresh}$, where R_{tamp}^{thresh} comes from solving Eq. (19). For a given value for R_{tamp} , Eq. (20) can be solved for α , after which the efficiency calculations of Eqs. (10)–(12) can be done after replacing R_{bare}^{thresh} with R_{tamp}^{thresh} .

The factor of two reduction in the critical radius associated with an infinitely thick tamper can be understood by examining Eq. (19). Suppose that the tamper is very thick, $R_{tamp} \gg R_{tamp}^{thresh}$. In this case, Eq. (19) reduces to

$$(R_{tamp}^{thresh}/d_{core}) \cot(R_{tamp}^{thresh}/d_{core}) = 1 - (\lambda_t^{tamp}/\lambda_t^{core}), \quad (22)$$

which is equivalent to Eq. (8) in Ref. 3. Now consider two subcases of this situation. If the tamper is a vacuum, we set $\lambda_t^{tamp} = \infty$, because a vacuum would have zero cross-section for neutron scattering. Equation (22) then becomes $(R_{tamp}^{thresh}/d_{core}) \cot(R_{tamp}^{thresh}/d_{core}) = -\infty$. This condition can be satisfied only by having $(R_{tamp}^{thresh}/d_{core})_{vacuum} = \pi$. The second subcase is that the tamper is thick and has a nonzero transport mean free path. For simplicity, assume that $\lambda_t^{core} \sim \lambda_t^{tamp}$; that is, the neutron-scattering properties of the tamper are much like those of the core. In this case, Eq. (22) becomes $(R_{tamp}^{thresh}/d_{core}) \cot(R_{tamp}^{thresh}/d_{core}) = 0$, which demands $(R_{tamp}^{thresh}/d_{core})_{thick} = \pi/2$. Thus, the critical radius is one-half of what it would be if no tamper were present. In reality, we do not expect such a dramatic effect.

IV. RESULTS

I used a spreadsheet to do the calculations that have been described.⁷ The spreadsheet was used to compute the results discussed in the following and shown in Fig. 1. After inputting data such as densities, atomic weights, cross-sections, the secondary neutron number, the average secondary-neutron energy, and values for E_f and γ , the user runs the Excel Goal Seek function to determine the threshold ($\alpha = 0$) bare critical mass, the yield of a bare core of chosen mass M_{core} , the threshold critical mass of a tampered core, and the yield and efficiency of a tampered core of the same mass. The tamper outer radius R_{tamp} has to be specified.

The Hiroshima *Little Boy* core comprised about 64 kg of enriched U-235 in a cylindrical configuration surrounded by a cylindrical tungsten-carbide tamper of diameter and length 13 inches, mass approximately 310 kg, and density 14.8 g/cm³.⁸ Tungsten has five naturally-occurring isotopes, W-180, 182, 183, 184, and 186, with abundances 0.0012,

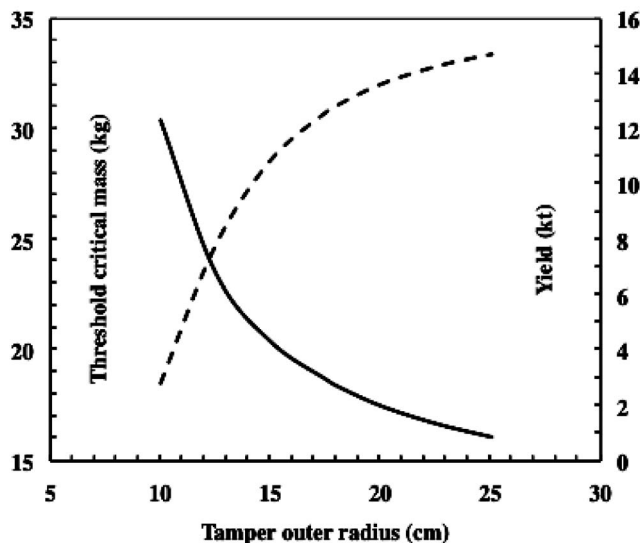


Fig. 1. Threshold tamped critical mass of a pure U-235 core (solid line) and yield of a 64-kg pure U-235 core (dashed line) as functions of the outer radius of a surrounding tamper of tungsten-carbide. The tamper material is assumed to have a density of 14.8 g/cm^3 , atomic weight of 195.84, and elastic-scattering cross-section of 6.587 bn. Yields are based on 17.6 kt/kg and $\gamma=2/3$.

0.265, 0.1431, 0.3064, and 0.2843, respectively. The Korea Atomic Energy Research Institute table-of-nuclides site gives elastic-scattering cross-sections for the four most abundant of these (in order of increasing weight) as 4.369, 3.914, 4.253, and 4.253 bn.⁹ If we neglect the small abundance of W-180, the abundance-weighted average of these cross-sections is 4.235 bn. Adding the 2.352 bn elastic-scattering cross-section for C-12 gives a total of 6.587 bn. The cross-sections must be added, not averaged, because we are considering the tungsten-carbide molecules to be single scattering centers of atomic weight equal to the sum of the individual atomic weights for tungsten and carbon, $183.84 + 12.00 = 195.84$. If we assume these values and take the 64 kg core to be spherical (radius 9.35 cm at a density of 18.71 g/cm^3) and surrounded by a spherical tungsten-carbide tamper of outer radius 18 cm (mass 311 kg), I find that the tamped threshold critical mass of U-235 in this configuration is 18.4 kg, about a 60% reduction from the bare threshold critical mass of 45.9 kg (the fissility parameters for uranium are described in the following). The predicted yield of a 64 kg core plus 311 kg tamper is about 12.9 kilotons (kt) for $\gamma=2/3$. This predicted yield is in surprisingly good agreement with the estimated $\sim 12 \text{ kt}$ yield of *Little Boy*.¹⁰ Some of this agreement is fortuitous in view of the approximations we have made (isotropic diffusion, neglect of inelastic scattering, neglect of fissions following second criticality, and tamper retardation effects) and the facts that the *Little Boy* core was not spherical or made of pure U-235, both of which have been assumed here. But it is encouraging that the model gives numbers in the correct ballpark.

The solid line in Fig. 1 shows how the threshold tamped critical mass for a U-235 core depends on the outer radius of a surrounding tungsten-carbide tamper. The dashed line shows the predicted yield for a 64 kg core of pure U-235. The mass of the tamper would be about 11 kg for an outer radius of 10 cm and just over 900 kg for an outer radius of 25 cm. Yields are based on $E_f=180 \text{ MeV}$, $\gamma=2/3$,

per-neutron kinetic energy 2 MeV, and an equivalence of $4.2 \times 10^{12} \text{ J/kt}$. The fissility parameters for U-235 are the same as those used in Ref. 1, $(\sigma_f, \sigma_{el}, \nu) = (1.235 \text{ bn}, 4.566 \text{ bn}, 2.637)$.

We might wonder why tungsten-carbide was used as a tamper material. Because one of the purposes of the tamper is to retard core expansion, dense tamper materials would presumably be preferable. It would seem that depleted uranium, which the Manhattan Project possessed in abundance, would be an ideal tamper material. Many reasons might be involved. Depleted uranium has a high spontaneous fission rate of about 675 per kilogram per second. Over the approximately 100 microseconds required to assemble the core of a Hiroshima gun-type bomb, a 300 kg tamper would have a significant probability of suffering a spontaneous fission and hence initiating a predetonation. Also, U-238 has a high inelastic-scattering cross-section for fast neutrons ($\sim 3 \text{ bn}$), which causes them to be slowed to the extent that they are very likely to be captured and removed from circulation. Former weapons designer Theodore Taylor has pointed out that beryllium is one of the best neutron reflectors known: its fission-spectrum averaged elastic scattering cross-section is about 2.7 bn, and its inelastic-scattering cross-section is only about 40 microbarns.¹¹

In summary, many aspects of the physics of fission-bomb cores are not mysterious and are understandable at the upper-undergraduate level and offer opportunities for students to see applications of basic physical principles to approximate models of these fascinating and complex devices.

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⁵As in Refs. 2 and 3, I neglect inelastic scattering, which has the effect of slowing neutrons to the point that they are unable to participate in causing fissions before the explosion is complete. I have not found a treatment of how to incorporate this effect and am unable to develop a convincing one myself. Another approximation is that weapons-grade uranium is usually not pure U-235; the neutron-absorbing effects of U-238 should be included, as well as the fact that U-235 has a small ($\sim 0.09 \text{ bn}$) radiative-capture cross-section for fission-energy neutrons. In Sec. IV, I model the *Little Boy* bomb as a U-235 core surrounded by a tungsten-carbide tamper. Tungsten isotopes also have significant inelastic-scattering cross-sections (2–3 bn) and smaller cross-sections for radiative capture (tens of millibarns) and production of protons, deuterons, and alpha-particles upon neutron bombardment (microbarns). For fission-energy neutrons, the vast majority of the carbon cross-section is for elastic scattering.

⁶I am grateful to a reviewer for pointing out this refinement.

⁷The spreadsheet is available at othello.alma.edu/~reed/Criticality.xls.

⁸nuclearweaponarchive.org/Nwfaq/Nfaq8.html.

⁹www.atom.kaeri.re.kr.

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